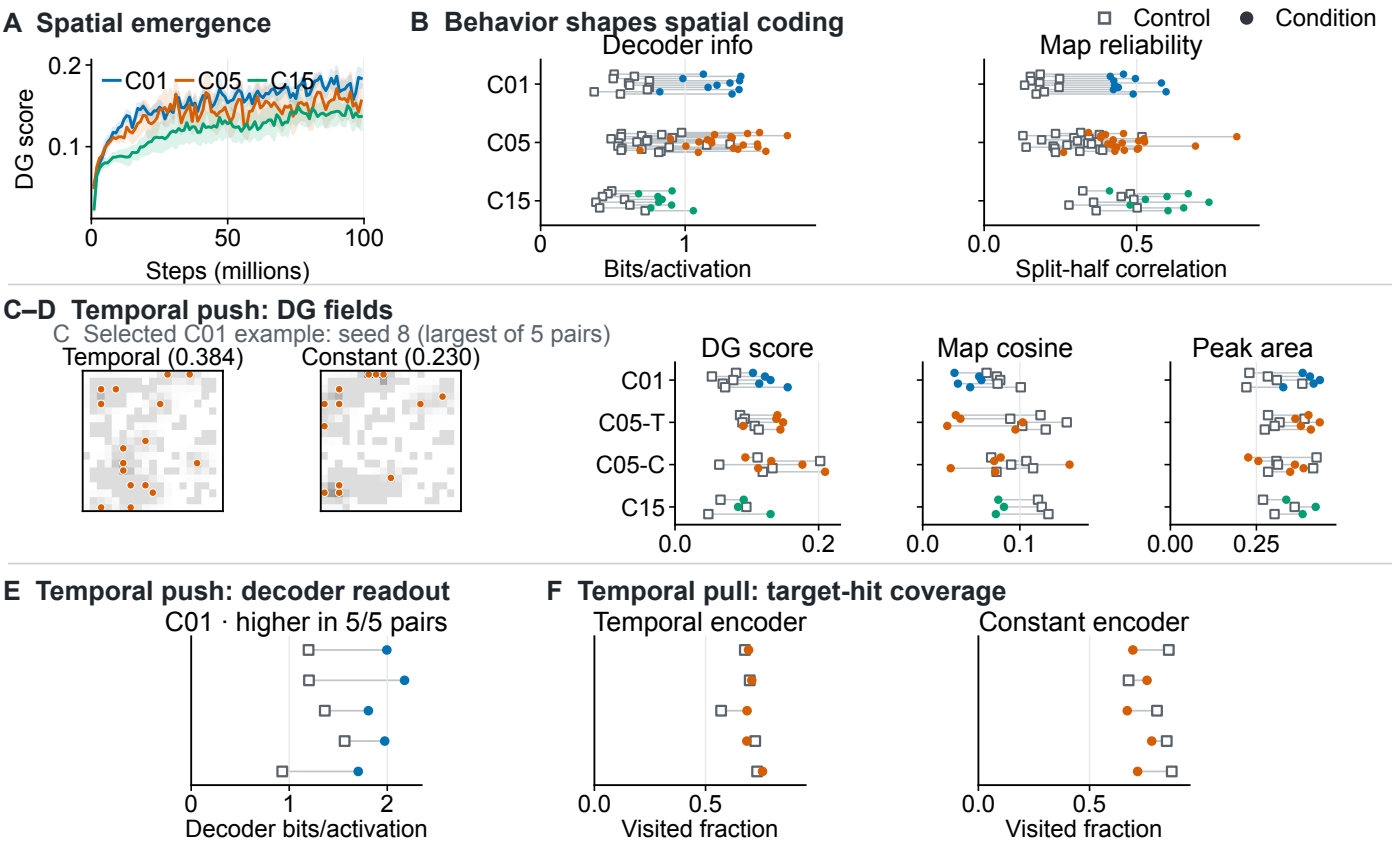

A Spatial emergence
DG score
0.2
0.1
C01
C05
C15
0
50
100
Steps (millions)
B Behavior shapes spatial coding
Control
Condition
Decoder info
C01
C05
C15
0
1
Bits/activation
Map reliability
0.0
0.5
Split-half correlation
C–D Temporal push: DG fields
C Selected C01 example: seed 8 (largest of 5 pairs)
Temporal (0.384)
Constant (0.230)
DG score
Map cosine
Peak area
C01
C05-T
C05-C
C15
0.0
0.2
0.0
0.1
0.00
0.25
E Temporal push: decoder readout
C01 · higher in 5/5 pairs
0
1
2
Decoder bits/activation
F Temporal pull: target-hit coverage
Temporal encoder
Constant encoder
0.0
0.5
Visited fraction
0.0
0.5
Visited fraction